Review intelligence for fit, usage, and VOC decisions

adiFit converts product reviews into structured signals for fit, comfort, design, use cases, customer segments, and actionable merchandising insights.

WHAT IT BUILDS

An AI review analysis tool that extracts fit, comfort, design, use-case, and VOC signals from product reviews.

HOW IT WORKS

Gemini structured output → embeddings/RAG → Streamlit dashboard → clustering & trend view.
Repo: github.com/HyeokyYun/26ACEProgram_TechDigitalOmni

KEY RESULT

24 demo reviews, 3 products, 5 customer segments, and 95.8% size-signal extraction accuracy.

CHALLENGE & SOLUTION

To reduce ungrounded size advice, the advisor answers only from retrieved review evidence and displays review_id citations with a coverage check.

LIMITATIONS & NEXT STEPS

The current demo uses curated sample reviews. The next step is connecting first-party reviews, returns, CS data, and product catalog signals.

Recorded product demo is submitted separately.

The accompanying video demonstrates the working prototype end to end within three minutes: fit advice, review-grounded evidence, merchandising dashboard, product comparison, segmentation, trend view, and validation.

DEMO SCENARIO

Samba OG fit question

Product: Samba OG
Usual size: 260
Width: wide
Question: “Can I buy my regular size?”

WHAT THE VIDEO SHOWS

Working AI flow

Review retrieval, LLM answer generation, cited evidence, dashboard charts, business action recommendations, review clusters, and validation metrics.

RUNTIME

Under 3 minutes

The video is provided as a separate attachment so the presentation remains lightweight and portable.

Reviews contain the signal, but the signal is buried.

Fit, width, comfort, design, use cases, and quality issues are mixed inside free-text reviews, making them hard for both shoppers and internal teams to interpret quickly.

CONSUMER + MERCHANDISER

Consumer friction

Shoppers need to manually scan reviews to answer questions such as “Will this fit true to size?” or “Is it comfortable for commuting?”

Operational friction

Fit complaints, durability concerns, delivery issues, and price resistance are difficult to categorize and turn into PDP actions.

Data challenge

Reviews are short, informal, and often contain positive feedback, negative feedback, and usage context in the same sentence.

AI opportunity

LLMs can structure review signals, while retrieval, aggregation, clustering, and trend views make those signals usable.

One analysis engine, two practical interfaces.

FOR CONSUMERS

Grounded Fit Advisor

Users provide their usual size, width context, and question. The app retrieves relevant reviews and generates size advice with cited evidence.

> “I usually wear 260 and have wide feet. Can I buy my regular size?”

FOR TEAMS

Merchandiser VOC Dashboard

The dashboard visualizes size signals, comfort, design, key VOC issues, customer segments, and monthly trends for product-page decisions.

> “Samba OG shows repeated wide-foot and size-up risk.”

A lightweight pipeline designed for a working prototype.

1

Review data

24 curated demo reviews across Ultraboost Light, Samba OG, and Adizero Adios Pro 3.

2

Structured extraction

Gemini extracts size, width, aspect sentiment, use cases, comfort, design/color, customer tags, and VOC issues.

3

Retrieval & clustering

Embeddings support RAG search, KMeans segmentation, PCA projection, and monthly tag aggregation.

4

Decision surfaces

Fit advisor, merchandising dashboard, product comparison, segment/trend view, and validation tab.

Why this shape

The pipeline is lightweight enough to run as a demo while preserving clear boundaries between extraction, retrieval, aggregation, segmentation, and UI.

What scales next

The same contracts can accept live reviews, larger vector indexes, product catalog metadata, and return/CS signals without changing the workflow.

Stack: Python, Streamlit, Gemini, Pydantic, Plotly, scikit-learn

Unstructured reviews become reusable product signals.

The schema keeps the LLM output consistent across retrieval, aggregation, visualization, and evaluation.

PYDANTIC CONTRACT

SIGNAL	EXAMPLES	USED FOR
Fit & size	runs_small, true_to_size, runs_large, width signal	Advisor, PDP guidance
Aspect sentiment	fit_size, comfort, quality, durability, delivery, design, value	VOC dashboard
Usage context	running, walking, commute, daily, gym, race	Customer segments
Comfort & design	cushioned, responsive, breathable, easy_to_style, gets_dirty	Messaging and product comparison
VOC issues	fit_risk, durability, delivery_packaging, price_resistance	Action recommendations

Answers are generated from retrieved review evidence, not general guesses.

INPUT

Product: Samba OG
Usual size: 260
Width: wide
Question: Can I buy my regular size?

RETRIEVAL

Top-k product reviews are retrieved by embedding similarity and passed to the LLM as the only allowed evidence.

OUTPUT

Recommendation, confidence, rationale, caveats, and cited review snippets are displayed in the app.

Recommended response pattern: size up for wide feet, with review_id citations and caveats for fit risk.

Review analysis is translated into merchandising actions.

Product-level insight

The dashboard shows review count, average rating, size verdict, width verdict, aspect sentiment, size-signal distribution, and top issues in one view.

VOC → action

Rules convert recurring signals into PDP and operations recommendations.

- Show half-size-up guidance when small-fit signals dominate.
- Recommend alternative models for wide-foot risk.
- Surface durability, delivery, or price resistance as operating VOC.

Clustering separates review patterns into usage segments.

Embeddings create clusters; structured tags explain the clusters in business language.

KMEANS + PCA

Comfort-driven Buyers

Reviews focused on cushioning, responsiveness, and long-wear comfort.

Wide-foot Fit Risk

Reviews focused on wide feet, tight fit, size-up needs, exchanges, and refund risk.

Performance Runners

Reviews focused on running, racing, record goals, responsiveness, and lightweight performance.

Post-purchase VOC Risk

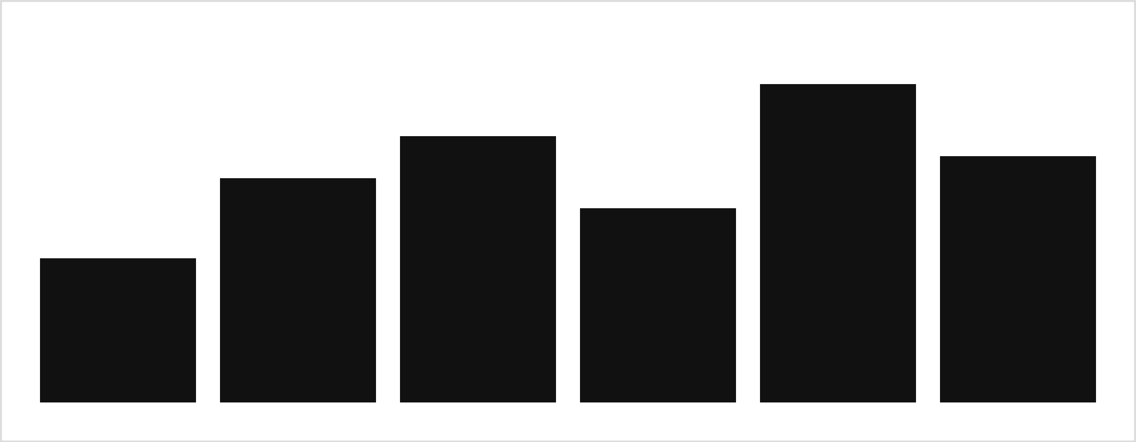
Reviews focused on durability, delivery/packaging, and price resistance after purchase.

At larger scale, the same approach can detect customer segment drift and emerging product issues over time.

Monthly VOC tags make issue movement visible.

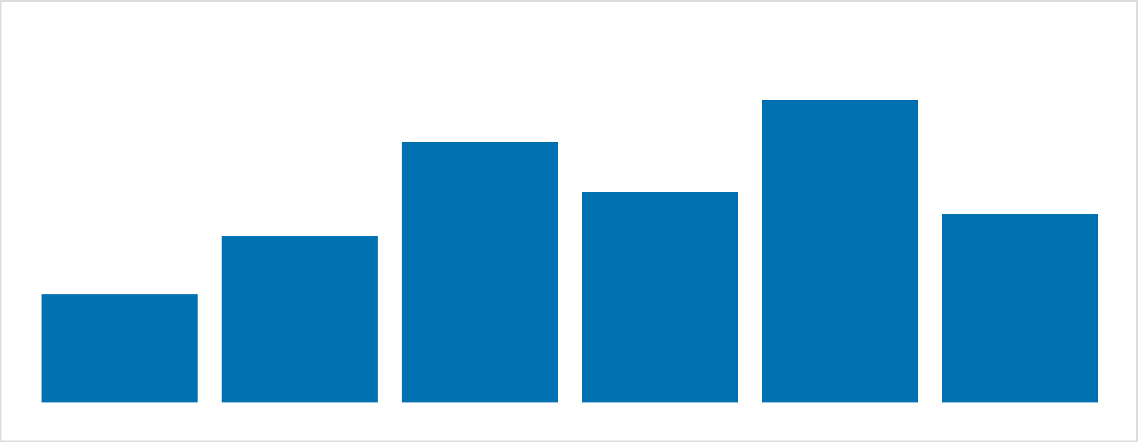
Issue trend

Fit risk, durability, delivery/packaging, price resistance, and comfort complaints are tracked by month.



Use-case trend

Running, daily, commute, race, and long-wear comfort signals can be monitored as seasonal demand patterns.



The prototype includes a lightweight validation layer.

24
REVIEW SAMPLES

3
PRODUCTS

95.8%
SIZE SIGNAL ACCURACY

100%
CITATION COVERAGE TESTS

Extraction validation

Manual gold labels check whether the model correctly extracts size_signal for each review. One ambiguous review is intentionally visible in the validation tab.

Grounding validation

Citation coverage checks whether every cited review_id exists in the retrieved evidence set. This is a grounding check, not a complete factuality benchmark.

The current evaluation is intentionally narrow and transparent: it proves the core loop works, while larger datasets are needed for full production-quality benchmarking.

From prototype to operational review intelligence.

DATA SCALE

Connect first-party review streams, return reasons, CS tickets, product catalog, and PDP analytics.

MODEL QUALITY

Expand evaluation beyond size signal into aspect sentiment, citation faithfulness, and segment stability.

BUSINESS LOOP

Trigger PDP copy updates, fit guidance, alternative recommendations, and VOC alerts from emerging signals.

STRUCTURED REVIEWS | GROUNDED ADVICE | CUSTOMER SEGMENTS | VOC TREND MONITORING